1 **(a)** $3 : 5 : 7$ **[B1]**

(b) 80 **[B2]**

If 0 scored, M1 for $\dfrac{240}{8+10+12} \times 10$ **[M1]**

[Total Marks: 3]

2 **(a)** $2 : 1$ **[B1]**

(b) $\dfrac{240}{2+3+5} \times (5-2)$ or $120 - 48$ **[M1]**

72 **[A1]**

[Total Marks: 3]

3 **(a)** M1 for 1.2 kg = 1200 g or 450 g = 0.45 kg

$3 : 8$ **[A1]**

(b) M1 for $3 + 4 + 5$ or 12 seen

140 **[A1]**

[Total Marks: 4]

4 **(a)** Total parts $= 2 + 3 + 5 = 10$ **[M1]**

Ben's share $= \dfrac{3}{10} \times 840 = 252$ **[A1]**

(b) $\dfrac{5}{10}$ **[M1]**

$\dfrac{1}{2}$ **[A1]**

[Total Marks: 4]

5 **(a)** $3 + 4 + 5 = 12$ **[M1]**

$\dfrac{5}{12} \times 120 = 50$ **[A1]**

(b) $\dfrac{3}{12} \times 120$ or 3×10 **[M1]**

30 **[A1]**

(c) $4 : 5$ **[B1]**

[Total Marks: 5]

6 **(a)** $45 : 120$ **[M1]**

$3 : 8$ **[A1]**

(b) $3.6 \times 100 = 360$ cm **[M1]**

$\dfrac{360}{2+3+7} \times 7$ **[M1]**

210 **[A1]**

[Total Marks: 5]

7 (a) $\dfrac{m+30}{f} = 3$ or $m + 30 = 3f$ **[B1]**

$\dfrac{m}{f+20} = 1$ or $m = f + 20$ **[B1]**

(b) M1 for substituting to eliminate one variable, e.g., $f + 20 + 30 = 3f$ or $m + 30 = 3(m - 20)$

M1 for correctly simplifying their equation to $2f = 50$ or $2m = 90$

$m = 45$, $f = 25$ **[A1]**

(c) 9 : 5 **[B1]**

[Total Marks: 6]

8 (a) 15 : 20 : 8 **[B2]**

If 0 scored, M1 for $3 \times 5 : 4 \times 5$ or $5 \times 4 : 2 \times 4$ **[M1]**

(b) 24 **[B2]**

If 0 scored, M1 for $\dfrac{129}{15+20+8} \times 8$ **[M1]**

(c) 6 **[B3]**

If 0 scored, M1 for $\dfrac{60-y}{24-y} = 3$ or $\dfrac{60-y}{24-y} = \dfrac{3}{1}$ **[M1]**

M1 for $60 - y = 72 - 3y$ (or equivalent linear equation) **[M1]**

[Total Marks: 7]

9 (a) 4.5×50000 or 225000 seen **[M1]**

2.25 **[A1]**

(b) Area scale is 1 cm^2 : 0.25 km^2 or map area ratio $1^2 : 50000^2$ **[M1]**

$14.2 \div 0.25$ or $14.2 \times 10^{10} \div 50000^2$ **[M1]**

56.8 **[A1]**

[Total Marks: 5]

10 (a) Actual distance $= 8.4 \times 250\,000 = 2\,100\,000$ cm **[M1]**

Distance in m $= 21\,000$ or distance in km $= \dfrac{2\,100\,000}{100\,000}$ **[M1]**

21 **[A1]**

(b) Area scale factor $= 1^2 : (250\,000)^2 = 1 : 6.25 \times 10^{10}$ **[M1]**

$35 \text{ km}^2 = 35 \times 10^{10} \text{ cm}^2$ **[M1]**

Map area $= \dfrac{35 \times 10^{10}}{6.25 \times 10^{10}} = 5.6$ **[A1]**

[Total Marks: 6]

11 (a) 16.4×250000 or 4100000 **[M1]**

41 **[A1]**

(b) Scale factor for area is $(250000)^2$ or 1 km $= 10^5$ cm so 1 km$^2 = 10^{10}$ cm^2 **[M1]**

$\dfrac{35 \times 10^{10}}{250000^2}$ or $\dfrac{35}{2.5^2}$ **[M1]**

5.6 **[A1]**

(c) 400000 **[B1]**

[Total Marks: 6]

12 (a) 29 **[B2]**

If 0 scored, M1 for $14.5 \times 1000 \times 100 \div 50000$ or equivalent **[M1]**

(b) 3.1 **[B2]**

If 0 scored, M1 for $12.4 \times \left(\dfrac{50000}{100000}\right)^2$ or 12.4×0.25 **[M1]**

(c) Calculates requires 360 g of flour and 225 g of butter **[B2]**

If B0 scored, M1 for $\dfrac{240}{8} \times 12$ or $\dfrac{150}{8} \times 12$ **[M1]**

No, because he needs 225 g of butter but only has 200 g **[B1]**

[Total Marks: 7]

13 (a) $16.8 \times 250\,000 = 4\,200\,000$ cm **[M1]**

42 **[A1]**

(b) Scale of area $= 1^2 : (2.5)^2$ OR 1 cm^2 represents 6.25 km^2 **[M1]**

8.96 **[A1]**

(c) 14 cm^2 represents 56 $km^2 \implies$ 1 cm^2 represents 4 km^2 **[M1]**

Linear ratio $= \sqrt{1}$ cm represents $\sqrt{4}$ km $\implies$ 1 cm to 2 km **[M1]**

$200\,000$ **[A1]**

[Total Marks: 7]

14 (a) M1 for $5 - 3 = 2$ parts

M1 for $84 \div 2$ or 42 seen

336 **[A1]**

(b) (i) M1 for 8.5×25000 or 212500 seen

2.125 **[A1]**

(ii) M1 for 1 cm represents 0.25 km or 1 km represents 4 cm

M1 for area scale factor 1 cm^2 represents 0.0625 km^2 or $3.5 \times 10^{10} \div 25000^2$

56 **[A1]**

[Total Marks: 8]

15 (a) Let total workers be $8x$. Males $= 5x$, Females $= 3x$. **[M1]**

Full-time males $= \frac{3}{5} \times 5x = 3x$. Part-time males $= \frac{2}{5} \times 5x = 2x$. **[M1]**

Full-time females $= \frac{4}{5} \times 3x = 2.4x$. Part-time females $= \frac{1}{5} \times 3x = 0.6x$. **[M1]**

Total full-time $= 3x + 2.4x = 5.4x$. Total part-time $= 2x + 0.6x = 2.6x$. Ratio is $5.4 : 2.6 = 27 : 13$. **[A1]**

(b) Difference $= 3x - 0.6x = 2.4x$ **[M1]**

$2.4x = 42 \implies x = 17.5$ **[M1]**

Total workers $= 8x = 8 \times 17.5 = 140$ **[A1]**

[Total Marks: 7]

16 (a) $5 - 3 = 2$ parts represent 42 workers **[M1]**

1 part is $\frac{42}{2} = 21$ **[M1]**

$8 \times 21 = 168$ **[A1]**

(b) Day shift workers: $5 \times 21 = 105$. Night shift workers: $3 \times 21 = 63$ **[M1]**

Day shift males: $\frac{3}{7} \times 105 = 45$. Day shift females: $105 - 45 = 60$ **[M1]**

Night shift males: $\frac{2}{9} \times 63 = 14$. Night shift females: $63 - 14 = 49$ **[M1]**

Total males $= 45 + 14 = 59$. Total females $= 60 + 49 = 109$. Ratio is 59 : 109 **[A1]**

[Total Marks: 7]

17 (a) Let the number of boys be $3x$ and girls be $5x$.

Forming equation: $\frac{3x - 4}{5x + 8} = \frac{1}{3}$ [M2]

(or M1 for either $3x - 4$ or $5x + 8$ seen in a ratio)

$9x - 12 = 5x + 8$ **[M1]**

15 **[A1]**

(b) Common ratio $A : B : C = 15 : 20 : 24$ **[M1]**

Difference in parts $24 - 15 = 9$, so 9 parts = 45 kg

$\implies$ 1 part = 5 kg **[M1]**

Total mass $= (15 + 20 + 24) \times 5 = 295$ **[A1]**

[Total Marks: 7]

18 **(a)** Alex : Ben = 15 : 20 and Ben : Chris = 20 : 24 **[M1]**

15 : 20 : 24 **[A1]**

(b) Difference in parts $= 24 - 15 = 9$ parts $\implies$ 9 parts = 16 200 **[M1]**

Total parts $= 15 + 20 + 24 = 59$ **[M1]**

$P = \dfrac{16\,200}{9} \times 59 = 106\,200$ **[A1]**

(c) Alex's share $= \dfrac{15}{59} \times 106\,200 = 27\,000$ **[B1]**

$27\,000 \times \left(1 + \dfrac{4.2}{100}\right)^5$ **[M1]**

33 167 **[A1]**

[Total Marks: 8]